Installing the programs

In order to participate in the workshop, you'll need to install these programs. You may already have some of them installed and if you did, please make sure that they are at least the minimum versions listed below. If not, you will need to update them.

Windows

1. R: Any version above . If you have used R before, you can confirm the version by running `R.version.string` in the Console.
2. RStudio: Any version above . If you have installed it before, check the current version by going to the menu "Help -> About RStudio".
3. Git: Select the "Click here to download" link. Git is used throughout many sessions in the workshop. When installing, it will ask for a selecting a "Text Editor" and while we won't be using this in the workshop, Git needs to know this information so choose Notepad.

MacOS

1. R: Any version above . If you have used R before, you can confirm the version by running `R.version.string` in the Console. If you use Homebrew (which we strongly recommend installing and using), installing R is as easy as opening a Terminal and running:

```
brew install --cask r
```

2. RStudio: Any version above . If you have installed it before, check the current version by going to the menu "Help -> About RStudio". With Homebrew:

```
brew install --cask rstudio
```

3. Git: Git is used throughout many sessions in the workshop. With Homebrew:

```
brew install git
```

Linux (Ubuntu)

1. R: Any version above . If you have used R before, you can confirm the version by running `R.version.string` in the Console.

```
sudo apt -y install r-base
```

2. RStudio: Any version above . If you have installed it before, check the current version by going to the menu "Help -> About RStudio".
3. Git: Git is used throughout many sessions in the workshop.

```
sudo apt install git
```

All these programs are required for the workshop, even Git. Git, which is a software program to formally manage versions of files, is used because of it's popularity and the amount of documentation available for it. Check out the online book Happy Git with R, especially the "Why Git" section, for an understanding on why we are teaching Git. Windows users tend to have more trouble with installing Git than macOS or Linux users. See the section on Installing Git for Windows for help.

> **Note**
>
> Some pictures may show a Git pane in RStudio, but you may not see it. If you haven't created or opened an RStudio R Project (which is taught in the introductory workshop), the Git pane *does not* show up. It only shows up in R Projects that use Git to track file changes.

> **Note**
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> *A note to those who have or use work laptops with restrictive administrative privileges*: You may encounter problems installing software due to administrative reasons (e.g. you don't have permission to install things). Even if you have issues installing or updating the latest version of R or RStudio, you will likely be able to continue with the workshop as long as you have the minimum version listed above for R and for RStudio. If you have versions of R and RStudio that are older than that, you may need to ask your IT department to update your software if you can't do this yourself. Unfortunately, Git is not a commonly used software for some organizations, so you may not have it installed and you *will* need to ask IT to install it. We *require* it for the workshop, so please make sure to give IT enough time to be able to install it for you prior to the workshop.

Once R, RStudio, and Git have been installed, open RStudio. *If you encounter any troubles during these pre-workshop tasks*, try as best as you can to complete the task and then let us know about the issues in the pre-workshop survey. If you continue having problems, indicate on the survey that you need help and we can try to book a quick video call to fix the problem. Otherwise, you can come to the workshop 15-20 minutes earlier to get help.

If you're unable to complete the setup procedure due to unfixable technical issues, you can use Posit Cloud (to use RStudio on the cloud) as a final solution in order to participate in the workshop. For help setting up Posit Cloud for this workshop, refer to the Posit Cloud setup guide.